

Zakaria Zair

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Montreal - QC

Education

Bachelor's Degree in Software Engineering from Polytechnique Montréal *2021 - 2025*
Data Science and AI - Award of Excellence

Work History

Full Stack Software Developer, CIMA+, Montréal *August 2025 - December 2025*

- Launched an end-to-end data analytics project for Canadian transit agencies to evaluate their prediction models against a standardized ETA benchmark
- Implemented a RESTful API and multithreaded SQL requests using Polars and PostgreSQL, improving I/O-bound performance by approximately 200%
- Developed multiple dashboards in React.js, enabling agencies to evaluate their own ETA algorithms
- Designed a Hexagonal architecture to optimize the workflows of the Python backend reducing average load times by 72%

Web Developer Intern, Global Affairs Canada, Ottawa *August 2022 - January 2024*

- Led the intranet website for a network of more than 10000 trade professionals in pure HTML, CSS, JavaScript and TypeScript
- Drove a persona analysis to search for relevant keywords and increase SEO impressions by 85%
- Automated social media posts subprocesses to reduce CI/CD overhead

Projects

Full Stack Software Developer, Desktop and Mobile Game 🎮 *January 2023 - May 2024*

- Spearheaded an online game project in TypeScript with a MANN stack in an Agile environment
- Implemented WebSocket and HTTP interfaces between deployed AWS & GCP servers and GitLab clients
- Designed game logic in TypeScript, account CRUD in MongoDB using SQL schemas and real-time synchronization with Socket.IO

Front End Developer, Health Application *February 2023 - Current*

- Developed web and iOS learning applications in React.js to provide useful medical information for doctors and health professionals, within the Comité De Développement Médical (CDM)
- Implemented in React.js for cross-platform builds with Node.js backend with Supabase integration with the use of AI agents
- Optimized client-server interactions to minimize the number of requests and resource loading at critical times

Software Architect, Website for Medical Students *January 2023 - Current*

- Drove a website in HTML, CSS, JavaScript gaining over 4000 registered users within the Comité De Développement Médical (CDM)
- Implemented a serverless Vercel API for a third-party website to fix CDM's privacy issues

Skills

Languages - TypeScript, JavaScript, CSS, HTML, Python, SQL, C#

Technologies - React.js, Node.js, Angular, Next.js, AWS, GCP, Azure, PostgreSQL, Docker, .NET

Tools - Git, GitLab, Zed Editor, Jira, Asana, ClickUp, Postman